GitHub outage prediction

PROPOSED PILOT

Early warning for bank access, integrations and Actions

1 Signals

Git + API probes
API and git ls-remote latency
DNS / TLS / proxy timing

Integration metrics
API quota, reset, call rate
Token renewal and failures

Actions + ARC metrics
Queue age and startup time
DEV / UAT / PRD capacity

Changes + network
Rollouts and secret rotation
Network route and IdP health

2 Collect & prepare

Metric collection
Prometheus + logs
Scrapers + probes
Data freshness checks

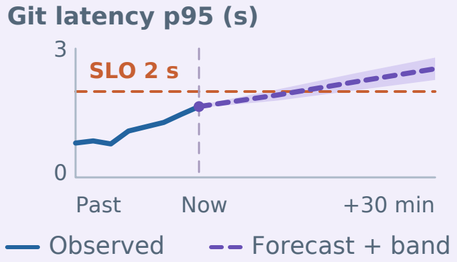
Every minute

Rolling features
5 / 15 / 60-min windows
p95, trend, seasonality
Per app / route / group

3 ML forecast

ML forecasting
Live inference on GKE

Forecast latency and demand
15 / 30 / 60-min horizons



Quota / expiry rules add evidence

4 Outage risk

Risk evaluation
API quota interruption
Git access degradation
Actions backlog

Warning gate
Fresh + persistent signal
Corroborating evidence
Time to potential impact

ILLUSTRATIVE WARNING
Git SLO at risk within 30 minutes

5 Respond

Grafana + alerts
Forecast window
Affected operation
Evidence and owner

On-call response
Check route / auth
Throttle polling / cache
Restore access / scale

Use an approved runbook

Async feature export

Validated model

HISTORY & LEARNING

Bank controls
Terraform-managed resources
Scoped secrets and IAM
Approved VPC / proxy egress

BigQuery history
Features + incidents
Event & ingestion times

Cloud Run training job
Train on earlier windows
Test on later incidents
Compare with baseline alerts

Versioned model
GCS model artifact
Approve model rollout

Pilot measures
Useful warning lead time
False alerts per service
Coverage of past incidents

Incident outcomes, impact onset and operator action

Proposed pilot. Forecasts and warning times are illustrative; validate useful lead time against actual incidents.

Artifactory outage prediction

PROPOSED PILOT

Early warning for artifact availability, storage and dependency downloads

1 Signals

Artifactory telemetry
OpenMetrics + request logs
Latency, errors, JVM health

Storage + database
Fresh storage / quota data
I/O, DB wait, connections

Artifact probes
Download time and success
Remote repository latency

Changes + network
Rollouts and build demand
Proxy / DNS / load balancer

2 Collect & prepare

Metric collection
Prometheus + logs
Scrapers + probes
Data freshness checks

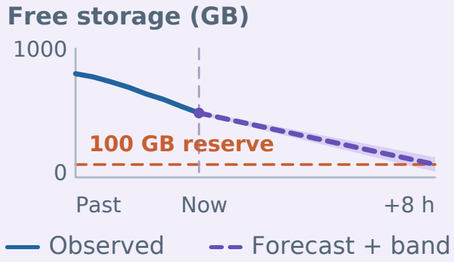
Every minute

Rolling features
5 / 15 / 60-min windows
p95, trend, seasonality
Per node / repo / route

3 ML forecast

ML forecasting
Live inference on GKE

Forecast storage and latency
30 min / 4 h / 8 h horizons



Capacity rules add evidence

4 Outage risk

Risk evaluation
Storage headroom risk
Download SLO breach
Database pressure

Warning gate
Fresh + persistent signal
Corroborating evidence
Time to potential impact

ILLUSTRATIVE WARNING
Storage reserve in about 8 hours

5 Respond

Grafana + alerts
Forecast window
Affected operation
Evidence and owner

On-call response
Check storage / DB
Add approved capacity
Approve retention

Use an approved runbook

HISTORY & LEARNING

Async feature export

Validated model

Bank controls
Terraform-managed resources
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Approved VPC / proxy egress

BigQuery history
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